Digital Transformation in the Age of Agentic AI

From Clay Tablets to Autonomous Agents: A Course on How Humans Adopt World-Changing Technologies

Format: 10 sessions | Case-method with applied analytical labs | MBA elective

A Note on Method

This is a course about executive judgment, not technical fluency. Students learn to recognize where a general-purpose technology sits on its adoption curve, judge which strategic frameworks survive contact with agentic AI, evaluate and cost deployment options, redesign organizations around the technology, and defend a transformation strategy to a board.

Course Description

Every MBA program now offers a course on digital transformation. Most begin with the internet era and end with a survey of today's AI tools — which means they equip students with a snapshot that will be obsolete soon. This course takes a radically different approach: it treats digital transformation as the newest chapter in humanity's oldest story — the invention, diffusion, and organizational absorption of transformative technology.

The course rests on three integrated arcs.

The first arc is historical. We begin not with ChatGPT but with clay tablets in Uruk, where writing — arguably the first information technology — was invented around 3400 BCE largely to keep track of stuff: sheep, beer, grain, and contracts. Writing was an accounting technology before it was literature; it restructured commerce, administration, and power. The Industrial Revolution offers the second great lesson: the technology was necessary but never sufficient. Britain industrialized first not because it alone had steam, but because of institutions, political stability, contract law, and a culture that dignified commerce and tinkering — and even then, median living standards fell for decades before the gains arrived. Electrification offers the third: factories owned electric motors for thirty years before productivity moved, because managers had bolted the new technology onto steam-era factory designs rather than reorganizing work around it. Students will learn to recognize this pattern — general-purpose technology arrives, incumbents retrofit, productivity disappoints, then organizational redesign unlocks the gains — because they are living through its fourth occurrence.

The second arc is analytical. Students master the foundational economics of the digital age — network effects and winner-take-all dynamics, switching costs and lock-in, platform and two-sided market strategy, the long tail, data network effects, information asymmetry and transaction costs, API and ecosystem strategy, and disruption theory — and then stress-test each concept against agentic AI. This is where the course breaks new ground: agentic AI doesn't merely illustrate these frameworks, it destabilizes them. If autonomous agents can migrate your data, learn a new interface instantly, and negotiate on your behalf, what happens to switching costs? If an AI agent does your shopping, who captures the value of a brand — and does Akerlof's lemons problem dissolve or migrate to the agent layer? If transaction costs collapse toward zero, Coase predicts firm boundaries shrink: is the "one-person unicorn" the logical endpoint? Students don't just learn the canon; they interrogate whether the canon survives.

The third arc is evaluative and organizational. Every analytical session pairs with hands-on work in the judgments an executive actually owns: costing an AI workload and defending build-versus-buy at a capital committee; mapping which competitive moats hold, erode, or migrate; decomposing real jobs into tasks and redesigning the workflow around human, agent, and hybrid execution; auditing an autonomous system for failure modes and accountability gaps; and placing governance guardrails at the right points in the risk-speed tradeoff. None of this requires a technical background, and none of it requires students to write code. The MBA who has costed, redesigned around, and stress-tested an agent will make fundamentally better capital-allocation and governance decisions than one who has only read about them.

Running through all three arcs is a question the technology history makes unavoidable, and most AI courses ignore: **who captures the gains, and when?** Electrification's productivity payoff arrived decades after the machines did, and the interval was not neutral for the people living through it. Students will confront the same question for agentic AI in Session 7, and it will follow them into the Blueprint.

The course culminates in an **Agentic Transformation Blueprint**: a board-ready plan for transforming a real organization, integrating historical judgment (where is this technology on its diffusion curve?), economic analysis (where will value be created and captured?), organizational design (what work gets redesigned, by whom, and what happens to the people doing it?), and governance.

Course Learning Objectives

Upon successful completion of this course, students will be able to:

CLO 1 — Historical Pattern Recognition. Analyze the recurring dynamics of transformative technology adoption — invention, diffusion, institutional enablement, the retrofit trap, distributional lag, and organizational redesign — across five millennia of general-purpose technologies (writing, steam, electrification, digital), and apply these patterns to assess where agentic AI sits on its adoption curve and what that position implies for the timing and sequencing of organizational investment.

CLO 2 — Strategic and Economic Analysis. Evaluate competitive situations in digital markets using the foundational frameworks of digital economics — network effects and winner-take-all dynamics, switching costs and lock-in, platform and ecosystem strategy, data network effects, information asymmetry, transaction cost theory, and disruption theory — and critique the continued validity of each framework under conditions of agentic AI, constructing reasoned arguments about which strategic moats survive, which erode, and which migrate to new layers of the value chain.

CLO 3 — Applied Agentic Evaluation. Evaluate, cost, and specify agentic AI deployments — assessing architectural and sourcing options (centralized versus embedded, single versus multi-agent, build versus buy, general versus specialized models), constructing credible unit-economics and cost-at-scale analyses, and writing deployment specifications that a technical team could act on and a capital committee could approve — without requiring a technical background or the production of code.

CLO 4 — Governance and Risk Judgment. Assess the security, ethical, regulatory, and organizational risks of deploying autonomous AI systems — including prompt injection, disinformation, accountability gaps, and compliance exposure under GDPR/HIPAA/EU AI Act-class regimes — and develop governance plans that place guardrails, testing strategies, escalation paths, and human oversight at appropriate points in the risk-speed tradeoff.

CLO 5 — Integrative Transformation Leadership. Synthesize historical judgment, economic analysis, evaluative capability, and governance design into a coherent, board-ready transformation strategy for a real organization — one that specifies where value will be created and captured, what organizational redesign (not retrofit) is required, how the workforce transition will be managed, how progress will be measured, and how internal resistance and the "last mile" from pilot to practice will be handled.

Assessment

Component	Weight	Primary CLOs
Analytical Lab Portfolio	30%	CLO 2, CLO 3, CLO 4
Case Participation	30%	CLO 1, CLO 2
Agentic Transformation Blueprint + Executive Presentation	40%	CLO 5 (integrating 1—4)

1. Analytical Lab Portfolio — 30%

A cumulative portfolio of five artifacts, each graded against the CLO it evidences. No artifact requires code, an API key, or paid model usage.

Lab Artifact	Session	CLO Measured
Workflow documentation ("Sumerian scribe" exercise)	1	CLO 1
Build-vs-buy memo with unit economics and cost-at-scale model	4	CLO 3
Moat migration map for a named incumbent	6	CLO 2
Task decomposition and workflow redesign, with workforce transition plan	7	CLO 3, CLO 5
Failure-mode and accountability audit + one-page governance plan	8—9	CLO 4

Evaluation standard: artifacts are assessed on the quality of reasoning and the defensibility of recommendations — students must justify sourcing and architectural choices (CLO 3), cost viability (CLO 3), competitive logic (CLO 2), and risk mitigations (CLO 4) in business terms a board would accept.

2. Case Participation — 30%

Case discussions are the principal venue for historical and analytical reasoning. Participation is assessed against three dimensions:

- **Pattern application (CLO 1):** Does the student connect the case to the adoption-cycle framework? (e.g., diagnosing Ford's Highland Park as redesign versus Boeing's MAX as retrofit-era governance failure.)

- **Framework rigor and critique (CLO 2):** Does the student apply the correct economic lens — and, in the agentic stress-test discussions (Sessions 3—6, 8), argue coherently about whether the framework survives? Three designated **position rounds** are individually graded mini-debates:
  - Session 5: "Agents destroy Apple's lock-in — defend or refute."
  - Session 6: "Agentic AI is sustaining, not disruptive, for SaaS incumbents — defend or refute."
  - Session 7: "The productivity gains from agentic AI will accrue to labor within a decade — defend or refute."
- **Risk and governance reasoning (CLO 4):** Contribution quality in the Boeing 737 MAX and eBay/trust discussions.

3. Agentic Transformation Blueprint and Executive Presentation — 40%

Required Section	CLO Demonstrated
Adoption-curve diagnosis: where the technology and industry sit; retrofit-trap risks specific to this organization	CLO 1
Value creation and capture analysis: which moats hold, erode, or migrate; competitive positioning under agentic conditions	CLO 2
Solution specification and economics: deployment model (centralized vs. embedded), sourcing decision, integration approach, cost-at-scale	CLO 3
Governance and risk plan: guardrail placement, testing strategy, compliance mapping, escalation paths, KPIs for autonomous systems	CLO 4
Organizational redesign and change roadmap: task-level redesign, workforce transition, last-mile plan, resistance management, sequencing, executive pitch	CLO 5

Evaluation: 25% written Blueprint (graded by section against the CLO rubric) + 15% executive-panel presentation and defense, in which panelists probe across all five objectives. A Blueprint that is economically sound but historically naïve, or analytically sharp but ungovernable, cannot earn distinction — mirroring the integrated judgment the course exists to build.

Coverage Summary

	Lab Portfolio (30%)	Case Participation (30%)	Blueprint + Presentation (40%)
CLO 1 Historical pattern recognition	●	●●	●●
CLO 2 Strategic/economic analysis & critique	●●	●●	●●
CLO 3 Applied agentic evaluation	●●		●●
CLO 4 Governance & risk judgment	●●	●	●●
CLO 5 Integrative transformation leadership	●		●●

(●● = primary measurement; ● = supporting evidence)

Session Plan

Session 1 — Writing, Sheep, and Beer: The First Information Technology.

Why writing emerged from accounting; technologies as tools of coordination and control; Lévi-Strauss's warning that information technology is also power.

> *Case:* "The British East India Company" (HBS) — information, distance, and control in the first multinational.
>
> *Lab:* Students document one organizational workflow the way a Sumerian scribe would — revealing what firms actually record and why.

Session 2 — Steam, Institutions, and the Adoption Gap.

The Industrial Revolution and why institutions beat inventions; the electrification lesson and the 30-year productivity lag; David and Brynjolfsson on general-purpose technologies. Introduces the distributional question that Session 7 takes up in full: gains arrive late and unevenly.

> *Case:* "The Dawn of the Age of Steam" or "Ford: The Highland Park Plant" (HBS) — reorganizing work around the technology.
>
> *Framework debrief:* the Retrofit Trap.

Session 3 — The Economics of Networks and Winner-Take-All.

Network effects, Metcalfe's logic and its limits, tipping, standards wars from VHS to UPI; does agentic AI have network effects, or just scale economies?

> *Case:* "Reliance Jio: Winning the Indian Telecom Battle" (Ivey/HBS).

Session 4 — Platforms, Ecosystems, and the Agent Interface.

Two-sided markets, chicken-and-egg launches, governance and take rates; MCP and agent protocols as the new platform battleground — when agents become the customers, who owns the interface, and who pays the toll? Treated as a question of platform power and bargaining position, not integration mechanics.

Case: "Amazon Web Services" (HBS) — turning internal capability into the world's platform.

Lab: **Build-vs-buy memo.** Cost an AI workload at scale and defend a sourcing recommendation — general-purpose versus specialized models, vendor versus in-house, and the unit economics that determine viability. Written for a capital committee.

Session 5 — Switching Costs, Lock-in, and Data Moats in an Agent World.

The classic typology of lock-in; data network effects and the Hagiu-Wright skepticism tests; the provocation: agents as universal switching-cost destroyers.

Case: "Apple Inc. in 2023" (HBS) — the deepest ecosystem lock-in ever built, and its agentic vulnerability.

Position round: "Agents destroy Apple's lock-in — defend or refute."

Session 6 — Disruption Theory Meets Agentic AI.

Christensen's mechanism, its critics, and its application to AI: incumbents' data and distribution advantages versus their resource-allocation blindness; is agentic AI sustaining for software incumbents or disruptive to them?

Case: "Netflix vs. Blockbuster" (HBS) paired with a current mini-case on legacy SaaS versus agent-native startups.

Lab: **Moat migration map.** Take one named incumbent and chart, with evidence, which of its defenses hold, which erode, and which migrate to the agent layer — then state what the firm should do about it.

Position round: "Agentic AI is sustaining, not disruptive, for SaaS incumbents — defend or refute."

Session 7 — Organizational Design for Agentic Work: Tasks, Roles, and Who Captures the Gains.

The longest and most consequential session of the course. Centralized versus embedded deployment reframed as an organizational question rather than an architectural one: center of excellence versus federated ownership, who holds the P&L, what happens to spans of control, and why the same technology produces Highland Park in one firm and an expensive pilot graveyard in another. The second half turns to the distributional question raised in Session 2 — task displacement versus task augmentation, the skills premium, wage effects, the timing gap between adoption and gains, and what leaders owe the workforce during that interval.

Case: "Klarna's AI Customer-Service Transformation" or "Moderna: AI in Action" (HBS/current) — the enterprise that reorganized around the technology — paired with a short reading on the electrification-era distributional lag.

Lab: **Task decomposition and workflow redesign.** Decompose a real role into constituent tasks; assign each to human, agent, or hybrid execution; redesign the workflow around the result; and attach a workforce transition plan naming what happens to the people currently doing the displaced tasks.

Position round: "The productivity gains from agentic AI will accrue to labor within a decade — defend or refute."

Session 8 — Trust, Asymmetry, and the New Transaction Costs.

Akerlof, Spence, and Coase in the agent economy: reputation systems for agents, agent-to-agent commerce, the lemons problem at machine speed, and the firm-boundary question — if transaction costs collapse, is the "one-person unicorn" the logical endpoint or a category error? Deepfakes and disinformation as trust attacks.

Case: "eBay: Building Trust Among Strangers" (HBS) paired with an agent-commerce vignette.

Lab begins: **Failure-mode and accountability audit.** Working from a provided specification for a deployed customer-facing agent, identify plausible failure modes including prompt injection, tool misuse, and silent degradation, and trace who is accountable when each one occurs.

Session 9 — Governance, Guardrails, and the Last Mile.

Why pilots succeed and deployments stall; change management and internal resistance; the risk-speed quadrant; regulatory frameworks (GDPR, HIPAA, the EU AI Act) as strategic posture rather than compliance checklist; KPIs for autonomous systems; and the board's question — who signs the risk register for a system that acts on its own?

Case: "The Boeing 737 MAX" (HBS) — automation, oversight, and organizational accountability as the cautionary template for agent governance.

Deliverable: One-page governance plan with guardrail placement, escalation paths, and testing strategy, completing the Session 8 audit.

Session 10 — The Blueprint: Leading the Fourth Retrofit.

Digital transformation maturity models extended to agentic readiness; synthesis across all three arcs; student presentations of Agentic Transformation Blueprints to a panel of executives; closing reflection — what the scribes of Uruk, the engineers of Highland Park, and today's transformation leaders share, and what each generation learned too late about who bore the cost of the transition.

Case: "DBS Bank: Becoming a Technology Company" (HBS/Ivey) — the completed transformation as benchmark.

Course Policies

Prerequisites. None. No technical background, coding experience, or prior AI coursework is assumed.

Materials and cost. Cases as assigned. LLM model access.

Blueprint subject selection. Students choose their own organization for the Agentic Transformation Blueprint, subject to instructor approval by Session 4. The organization must be real, must be one the student can research substantively, and must not be the subject of a major deliverable the student is submitting in another course this term.

Use of generative AI. Students may use AI tools for research with disclosure. The analyses and judgments — which framework applies, which moat erodes, what the workforce transition costs, what the board

should approve — must be the student's own, and will be probed directly in class discussion and in the Blueprint defense.

What Makes This Course Unlike Any Other

No existing MBA offering spans five millennia of technology adoption, subjects the digital-strategy canon to the agentic stress test, forces the distributional question that every previous general-purpose technology posed, and requires every student to cost, redesign around, and govern an autonomous system before recommending it to a board. Graduates leave with something rarer than tool fluency: pattern recognition across technological revolutions — the judgment to know where we are in the cycle, which strategic frameworks still hold, who will capture the gains, and how to lead organizations through the reorganization that every general-purpose technology ultimately demands.